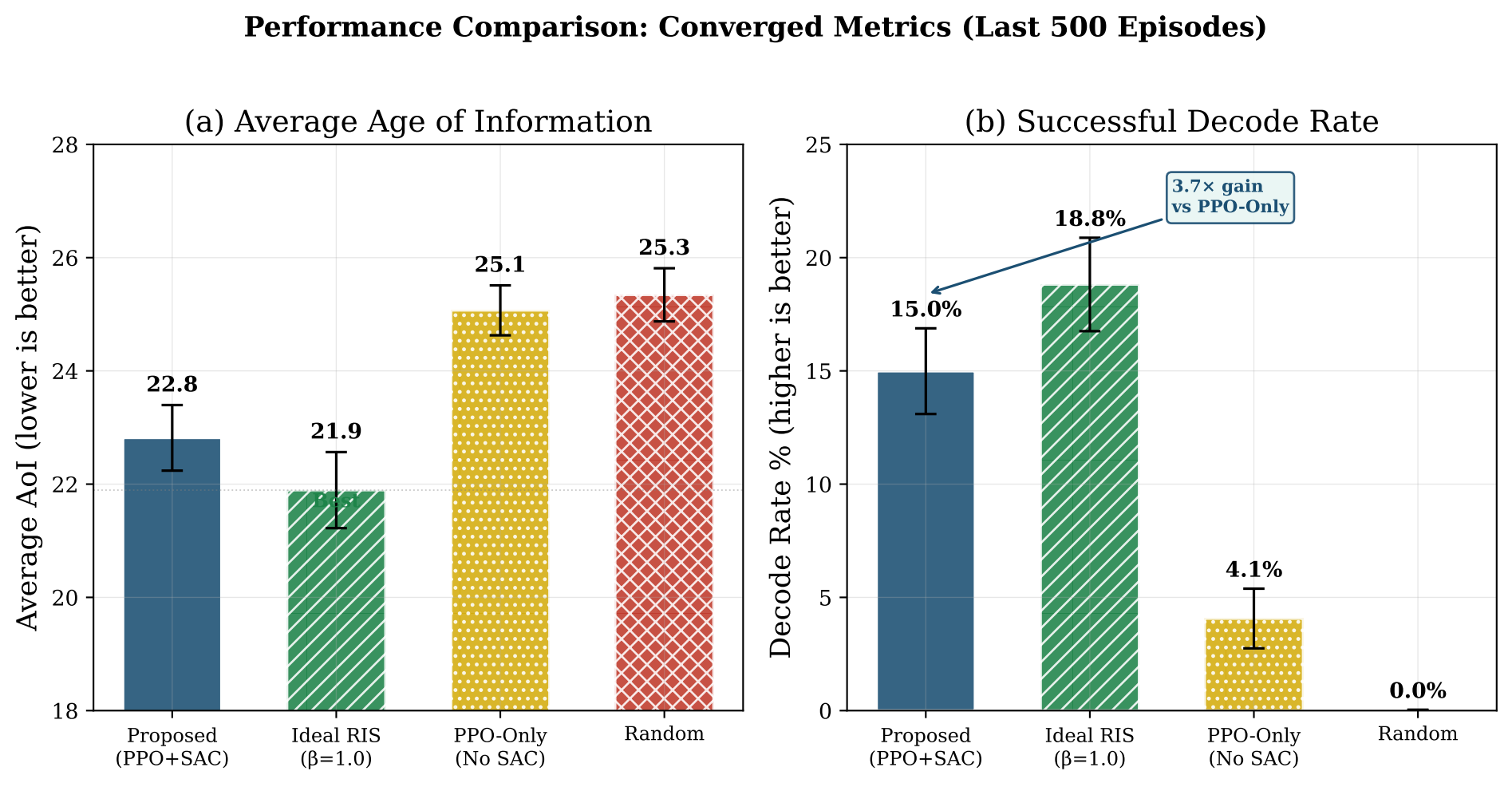

Performance Comparison: Converged Metrics (Last 500 Episodes)
(a) Average Age of Information
Average AoI (lower is better)
28
26
24
22
20
18
22.8
21.9
25.1
25.3
Proposed (PPO+SAC)
Ideal RIS (β=1.0)
PPO-Only (No SAC)
Random
(b) Successful Decode Rate
Decode Rate % (higher is better)
25
20
15
10
5
0
3.7× gain vs PPO-Only
15.0%
18.8%
4.1%
0.0%
Proposed (PPO+SAC)
Ideal RIS (β=1.0)
PPO-Only (No SAC)
Random